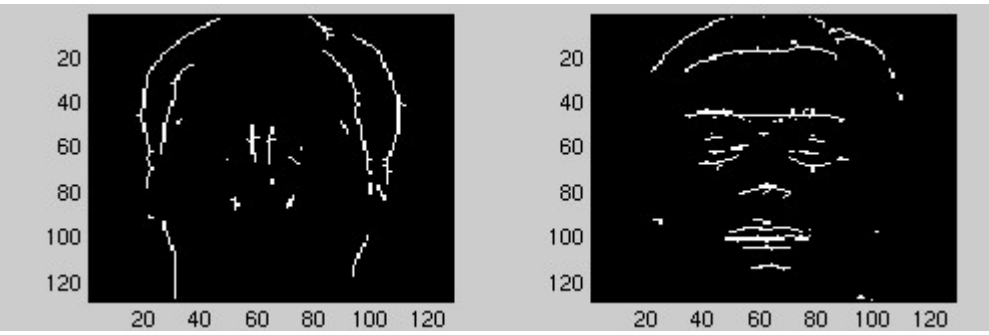
1. Given the original grayscale on the left, we apply two different convolutional filters to obtain images (a) and a(b). Explain the filters and give filter coefficients as a 3x3 matrix.




2. Answer the questions below.
    a. Give 2 benefits of using convolutional layers instead of fully connected ones for visual tasks.
    b. You are given a content image Xc and style image Xs. You would like to apply neural style transfer to obtain an output image Y, with the content of Xc and the style of Xs. You are told that you need a pretrained VGG-16 network to do his. What is the function of this pretrained network? Explain in detail.
    c. Explain the benefits of using batch normalization.
3. Answer the questions below.
    a. You have a CNN classifier. For each layer, calculate the number of weights, number of bias and the size of the activation maps. Give your answers as a multiplication of numbers, as shown in the table below.
    CONV-K-N: N filters with size KxK padding=0 stride=1
    POOL-K: KxK pooling, stride=K, padding=0
    FC-N: fully-connected layer with N neurons

| Layer | Activation map dim. | # of weights | # of biases |
|---|---|---|---|
| INPUT | 128x128x3 | 0 | 0 |
| CONV-9-32 | | | |
| POOL-2 | | | |
| CONV-5-64 | | | |
| POOL-2 | | | |
| CONV-5-64 | | | |
| POOL-2 | | | |
| FC-3 | | | |

    b. Following the last FC-3 layer of your network, what activation must be applied? Given a vector a=(0.3, 0.3, 0.3), what is the result of using your activation on this vector?
    c. You find online that the exact same network has already been trained on 1,000,000 samples of similar classification task. What is the name of the method that could we re-use the pretrained weights for the task at hand?
    d. What are the hyperparameters to tune for this method?